

EXPT NO:7 IMPLEMENT ADVANCED CHART TYPES USING TABLEAU AND POWER BI

DATE : 14.08.2026

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What is data integration in the context of Business Intelligence (BI)?

Data integration in Business Intelligence is the process of combining data from different sources into a unified and consistent view for analysis and reporting. It allows organizations to bring together sales, targets, customer, and other business data so that relationships and comparisons can be analyzed in one dashboard. In this experiment, retail sales data and regional sales target data are integrated to compare actual performance with planned targets.

2. Explain the difference between a join and a data blend in Tableau.

A join in Tableau combines tables at the data-source level using a common field and produces a combined table based on a specified join type. A data blend combines data from separate data sources at the visualization level using a linking field. Joins are generally used when the data can be combined directly, while blending is useful when the sources must remain separate or cannot be joined directly.

3. What are relationships and cardinality (e.g., one-to-many, many-to-many) in Power BI's data model?

A relationship in Power BI defines how tables are connected through common fields. Cardinality describes how records in one table correspond to records in another table. One-to-many means one record in one table can relate to many records in another table; many-to-many means multiple records on both sides can correspond to multiple records on the other side. Correct relationships and cardinality help Power BI calculate and filter data accurately.

4. Why is data quality (e.g., consistent naming conventions, data types) crucial when integrating data from different sources?

Data quality is crucial because inconsistent names, incorrect data types, duplicates, and missing values can produce incorrect joins, calculations, and visualizations. For example, 'N. America' and 'North America' may represent the same region but will not match unless the naming is standardized. Correct and consistent data improves the reliability and accuracy of the integrated dashboard.

5. Name two common challenges encountered during data integration.

Two common challenges are mismatched values or naming conventions between sources and differences in data types or granularity. Other possible challenges include missing values, duplicate records, incompatible formats, and incorrect relationships.

IN-LAB EXERCISE:

OBJECTIVE:

To connect, combine, and model data from multiple sources in Tableau and Power BI and create a unified, comprehensive dashboard for holistic analysis.

PROCEDURE:

1. Data Collection:

Import two datasets:

- Dataset 1: Sales Data (CSV) → Order ID, Order Date, Region, Sales Amount
- Dataset 2: Regional Sales Targets (Excel) → Region, Sales Target, Target Year

2. Data Preparation and Transformation:

- Clean datasets (e.g., standardize "N. America" → "North America").
- Ensure correct data types (numeric for Sales, date for Order Date).
- Handle missing values appropriately.

3. Data Integration and Modeling:

- Tableau: Create a join/relationship on the Region field in the Data Source pane.
- Power BI: Use the Model view to create a relationship on Region; define cardinality (e.g., many-to-one).

4. Create Calculated Fields for Comparison:

- Target Achievement % = $(\text{SUM}(\text{Sales}) / \text{SUM}(\text{Sales Target})) * 100$
- Sales vs. Target Variance = $\text{SUM}(\text{Sales}) - \text{SUM}(\text{Sales Target})$

5. Design a Comprehensive Dashboard:

- Combine visuals from both sources.
- Add KPI cards: Total Sales, Overall Target Achievement %.
- Use gauge/bullet charts for region-wise target performance.
- Combined bar chart: Actual Sales vs. Target by Region.
- Map visualization: Sales vs. Target Variance by region.

6. Validate Data Integrity:

- Cross-check joined data for accuracy.
- Ensure dashboard filters correctly update visuals.

7. Publishing and Sharing:

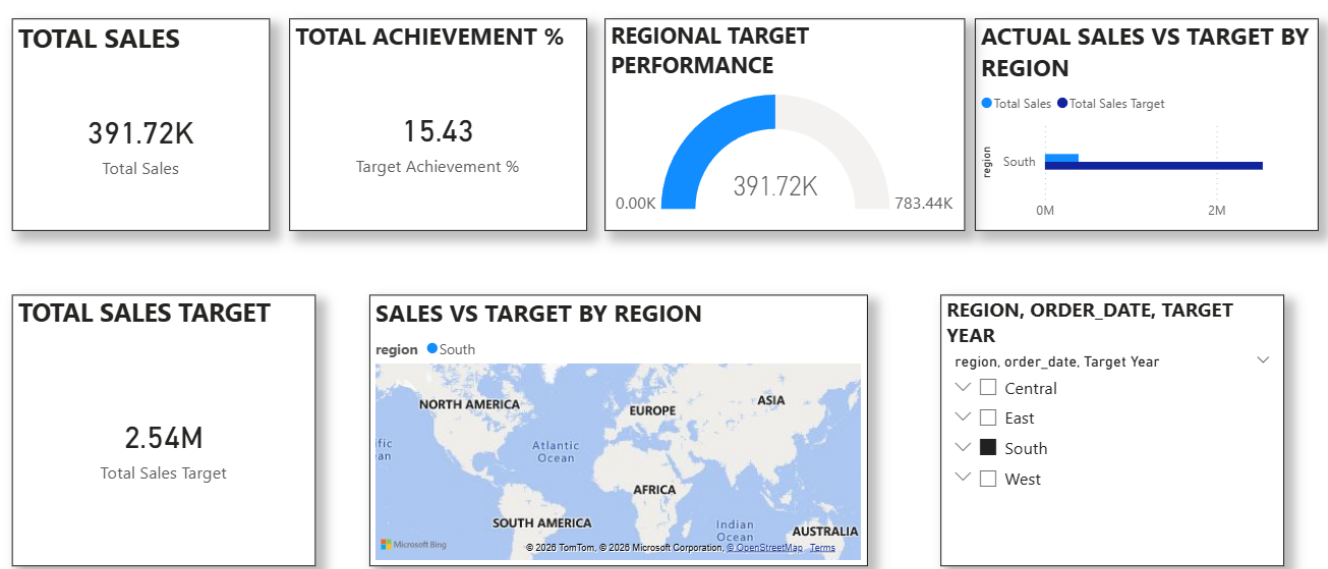
- Tableau: Publish to Tableau Public or Server.
- Power BI: Publish to Power BI Service and share.

Experiment Datasets:

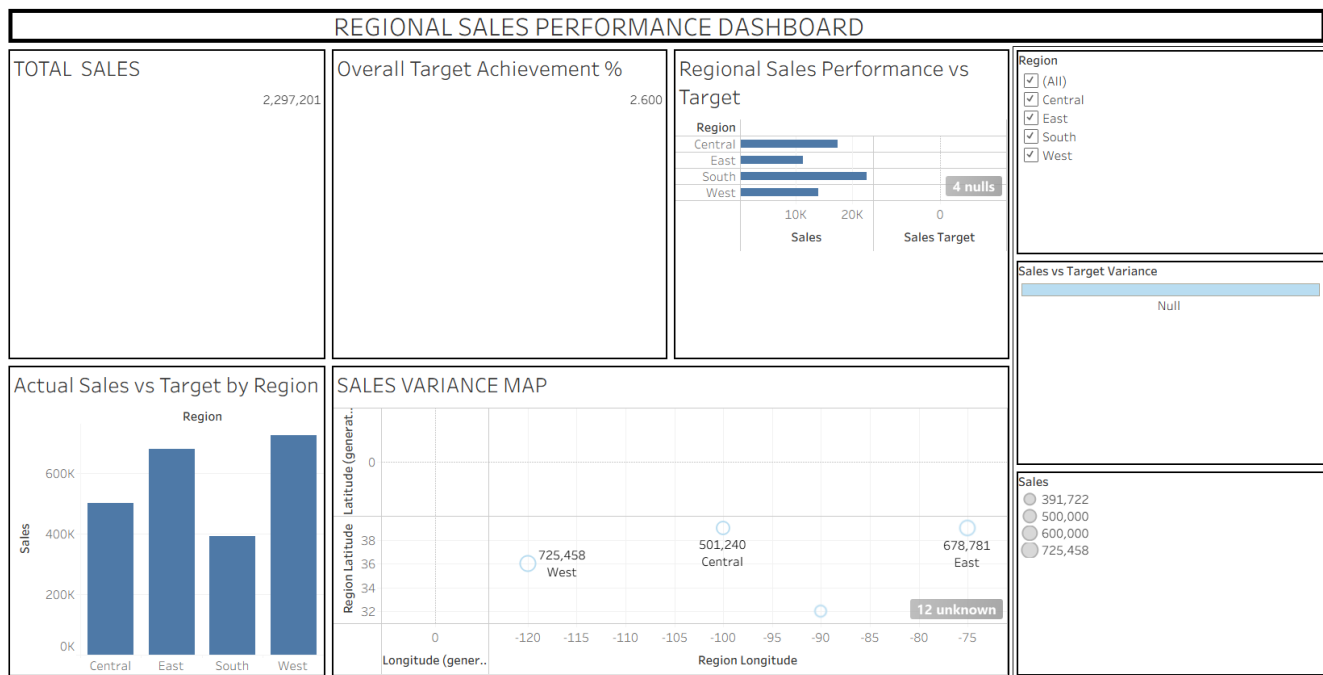
- Dataset 1: Retail Sales Data (CSV) → Order ID, Order Date, Region, Sales
- Dataset 2: Regional Sales Targets (Excel) → Region, Sales Target, Target Year

SCREENSHOT OF POWER BI

REGIONAL SALES PERFORMANCE DASHBOARD



SCREENSHOT OF TABLEAU



POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. How did integrating the sales targets data provide a more complete analytical picture than using the sales data alone?

Integrating sales targets with sales data provides a more complete analytical picture because actual sales can be compared with the planned target. Sales data alone shows how much was sold, but the target data shows whether that performance was sufficient. The integrated view supports KPI analysis, target achievement percentage, variance analysis, and region-wise performance comparison.

2. Describe the type of join or relationship you used to combine the two datasets and explain why it was the appropriate choice.

The datasets were combined using the Region field. In Tableau, a join or relationship can be created on Region, while in Power BI a relationship can be created on Region with suitable cardinality, such as many-to-one. This is appropriate because the sales records contain regional sales transactions while the target dataset provides the corresponding regional target. It allows actual sales to be compared with the target for each region.

3. What challenges, if any, did you face regarding data granularity or mismatched keys (e.g., different region names) between the two sources?

A possible challenge was mismatched region names or values between the two sources, such as 'N. America' and 'North America'. This was handled by standardizing the region names before integration. Differences in

data granularity can also cause duplication or incorrect totals, so the fields and relationship must be checked carefully before creating calculations and visuals.

4. How can creating variance calculations (e.g., Sales vs. Target) help a business manager make better decisions?

Variance calculations such as Sales vs. Target show whether actual performance is above or below the planned target. A positive variance indicates that sales exceeded the target, while a negative variance indicates that the target was not achieved. Managers can use this information to identify strong or weak regions, investigate performance gaps, allocate resources, and take corrective action.

5. What other data source (e.g., marketing campaign data, customer demographics) could you integrate to further enrich this analysis, and what new insights might it provide?

Marketing campaign data could be integrated to study whether promotions or campaigns influenced regional sales performance. Customer demographic data could reveal which customer groups contribute most to sales. Other useful sources include product data, inventory data, and customer feedback. These additional sources could provide insights into campaign effectiveness, customer segments, product demand, inventory planning, and reasons for regional performance differences.

RESULT:

The retail sales and regional targets datasets were successfully connected, cleaned, and integrated in Tableau and Power BI. Relationships between the sources created a unified data model. The dashboard visualized sales performance against targets, enabling variance analysis and providing insights into regional performance not possible with a single data source.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	

Faculty Signature